# Trainings

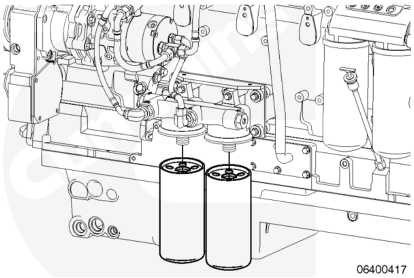
## General Information

**Note : This procedure applies to Stage 2 fuel filters on engines with electronically actuated injectors only.**

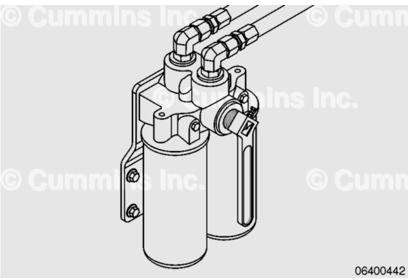
Use the following procedure for engines with mechanically actuated injectors. Refer to Procedure 006-015 in Section 6. (/qs3/pubsys2/xml/en/procedures/20/20-006-015-tr.html)

Use the following procedure for Stage 1 fuel filters. Refer to Procedure 006-075 in Section 6. (/qs3/pubsys2/xml/en/procedures/20/20-006-075-tr.html)


The Stage 2 filter head is mounted on the engine and is vibration isolated with vibration dampers. There is an option for the Stage 2 filter to be remote mounted, depending on the original equipment manufacturer (OEM).




The Marine Classed applications use duplex remote mounted Stage 2 filters to allow the filters to be changed while the engine is operating at idle.




## Preparatory Steps

### with Electronically Actuated Injector

**⚠ WARNING ⚠**

**Depending on the circumstance, fuel is flammable. When inspecting or performing service or repairs on the fuel system, to reduce the possibility of fire and resulting severe personal injury, death, or property damage, never smoke or allow sparks or flames (such as pilot lights, electrical switches, or welding equipment) in the work area.**


### Industrial and Power Generation Applications

- The Stage 2 filtration consists of an engine-mounted vibration-isolated dual filter head. Both filters **must** be used during normal engine operation.
- Shut the engine down.
- Close the fuel supply shutoff valve.
- Change the Stage 1 fuel filters as needed. Refer to Procedure 006-075 in Section 6. (/qs3/pubsys2/xml/en/procedures/20/20-006-075-tr.html)

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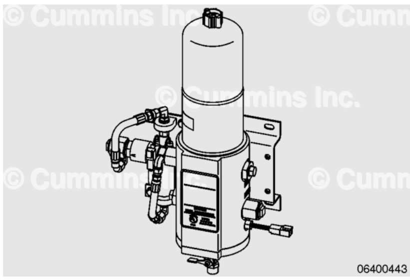
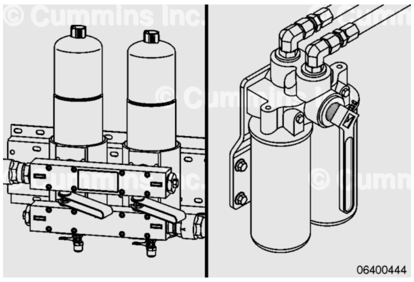



### Marine Non-Classed Applications

- The Stage 2 filtration consists of a vibration-isolated dual filter head that can be either engine-mounted or remote-mounted. Both filters **must** be used during normal engine operation.
- Shut the engine down.
- Close the fuel supply shutoff valve.
- Change the Stage 1 fuel filters as needed. Refer to Procedure 006-075 in Section 6. (/qs3/pubsys2/xml/en/procedures/20/20-006-075-tr.html)

**⚠ WARNING ⚠**

**Depending on the circumstance, fuel is flammable. When inspecting or performing service or repairs on the fuel system, to reduce the possibility of fire and resulting severe personal injury, death, or property damage, never smoke or allow sparks or flames (such as pilot lights, electrical switches, or welding equipment) in the work area.**




**Note :** This procedure explains Classed fuel system filter changes with the engine operating at idle. If the engine is Classed, and the intent is to change filters with the engine shut down, follow the Non-Classed procedure.

Marine Classed Applications

- The Stage 2 filtration consists of a remote-mounted duplex filter head. Both filter elements **must** be used during normal engine operation.
- Change the Stage 1 fuel filters as needed. Refer to Procedure 006-075 in Section 6. (/qs3/pubsys2/xml/en/procedures/20/20-006-075-tr.html)

With the engine operating at idle, move the selected valve handle to the OFF position (depending on which Stage 1 filter is to be changed.)

The Stage 2 duplex filter is designed to allow one of the filters to be changed while the engine is in operation at reduced capacity or at idle. To accomplish this, move the handle ninety degrees to the right or left, depending on the filter being changed. The valve has a “T” on its face to indicate which filter is in use. If the handle is moved to the left, the right filter can be changed. If the handle is moved to the right, the left filter can be changed.

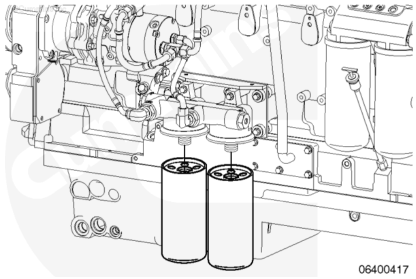
With the engine operating at idle, move the selected valve handle to the right or left position (depending on which Stage 2 filter is to be changed).

# Remove

## with Electronically Actuated Injector

**⚠ WARNING ⚠**

**Depending on the circumstance, fuel is flammable. When inspecting or performing service or repairs on the fuel system, to reduce the possibility of fire and resulting severe personal injury, death, or property damage, never smoke or allow sparks or flames (such as pilot lights, electrical switches, or welding equipment) in the work area.**



**⚠ CAUTION ⚠**

**Do not spill or drain fuel into the bilge area when disconnecting or removing fuel lines, replacing filters, and priming the fuel system. Do not drop or throw filter elements into the bilge area. The fuel must be disposed of in accordance with local environmental regulations.**

**Note :** There are two different filter heads used on the QSK19 Stage 2 filter. They can be either engine-mounted or remote-mounted.

On the dual filter head, remove the fuel filters with fuel filter wrench, Part Number 3400158, or equivalent.

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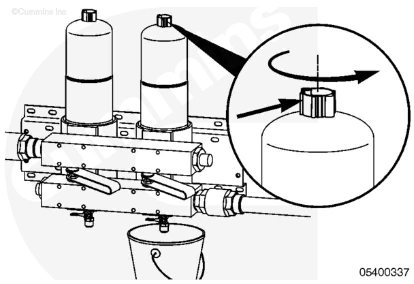


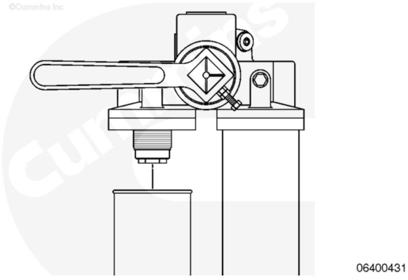
**⚠ CAUTION ⚠**

**Do not spill or drain fuel into the bilge area when disconnecting or removing fuel lines, replacing filters, and priming the fuel system. Do not drop or throw filter elements into the bilge area. The fuel must be disposed of in accordance with local environmental regulations.**

**Note :** This procedure explains the Classed fuel system filter changes with the engine operating at idle. If the engine is Classed, and the intent is to change filters with the engine shut down, follow the Non-Classed procedure.

### Marine Classed Applications

- Once the selected Stage 2 filter flow has been shut off, use filter wrench, Part Number 3400158, or equivalent, to remove the filter.



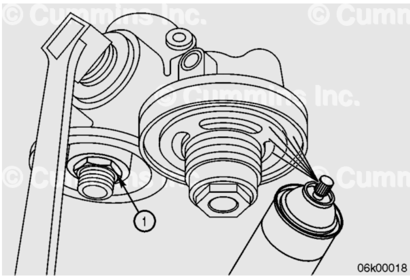

# Clean

## with Electronically Actuated Injector

Check that the thread adapter sealing ring (1) has been removed.

Spray the gasket mating surface of the filter head with cleaning solvent.

Allow the gasket mating surface of the filter head to air dry.



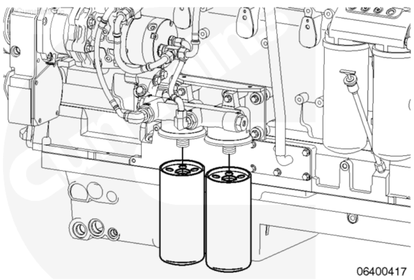

# Install

## with Electronically Actuated Injector

**Note : Do not fill the dual Stage 2 filters with fuel.**

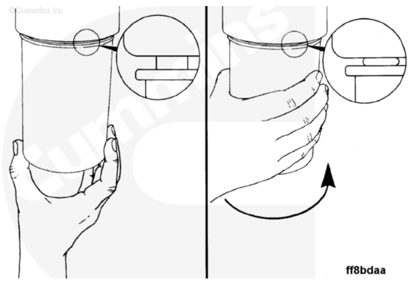
Lubricate the filter o-rings with clean engine oil.

Install the Stage 2 fuel filters on the dual filter head.




Turn the filter until the gasket touches the surface of the filter head.

Tighten the filter an additional ½ to ¾ of a turn after the gasket touches the filter head surface.

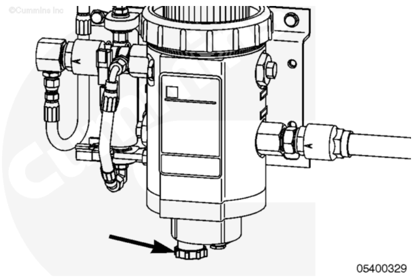


# Prime

## with Electronically Actuated Injector

**⚠ CAUTION ⚠**

**Do not spill or drain fuel onto the ground when disconnecting or removing fuel lines, replacing filters, and priming the fuel system. Do not drop or throw filter elements onto the ground. The fuel must be disposed of in accordance with local environmental regulations.**




### Industrial and Power Generation Applications

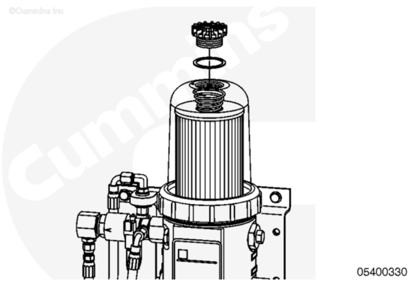
- Stage 1 and Stage 2 filters **must** be primed prior to starting the engine.
- Procedure for priming Stage 1 and Stage 2 filtration is covered in the same steps, as follows.

Verify the drain valve on the Stage 1 filter is closed at the base of the filter.

Close the fuel supply shutoff valve.

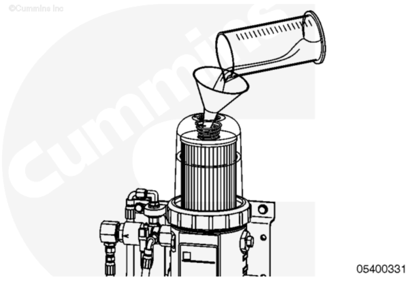
If the Stage 1 filters have drained, they **must** be filled prior to priming the Stage 2 filters.

Remove the vent cap from the top of the filter.




**⚠ CAUTION ⚠**

**Do not spill or drain fuel onto the ground when disconnecting or removing fuel lines, replacing filters, and priming the fuel system. Do not drop or throw filter elements onto the ground. The fuel must be disposed of in accordance with local environmental regulations.**



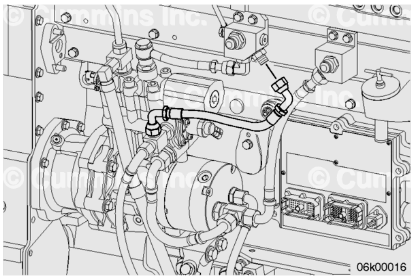

Fill the Stage 1 fuel filter with clean diesel fuel.

Install the vent cap.

Tighten the vent cap by hand **only**.

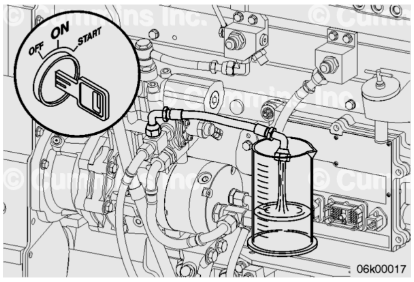
Open the fuel supply shutoff valve.

Remove the air bleed line from the drain manifold block and place it into a collection container



Turn the keyswitch ON and observe fuel exiting the air bleed line.

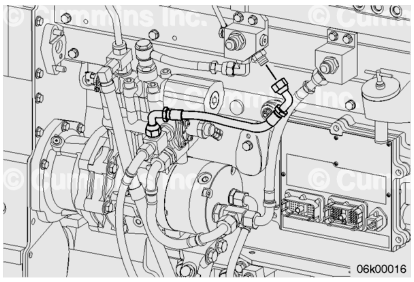
When a steady stream of fuel is observed, turn the keyswitch OFF. If the priming pump stops operating before steady stream of fuel is observed, turn the keyswitch OFF, wait 30 seconds, and turn the keyswitch ON.



Repeat as necessary to observe a steady stream of fuel.

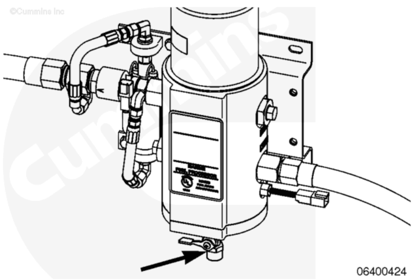
Install an air bleed line to drain the manifold block.

**Torque Value:** 45 n•m [ 33 ft-lb ]



**⚠ CAUTION ⚠**

**Do not spill or drain fuel into the bilge area when disconnecting or removing fuel lines, replacing filters, and priming the fuel system. Do not drop or throw filter elements into the bilge area. The fuel must be disposed of in accordance with local environmental regulations.**



## Marine Non-Classed Applications

- Stage 1 and Stage 2 filters **must** be primed prior to starting the engine.
- Procedure for priming Stage 1 and Stage 2 filtration is covered in the same steps, as follows.

Verify the drain valve is closed at the base of the Stage 1 filter.

Close the fuel supply shutoff valve.

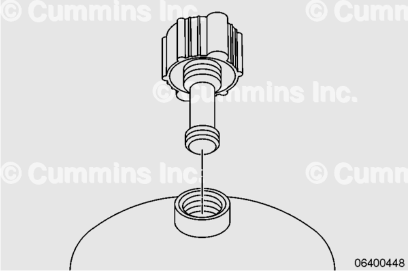
If the stage 1 filters have drained, they **must** be filled prior to priming the Stage 2 filters.

Unscrew the threads of the vent cap.

Lift the vent cap until the retainer shaft catches.

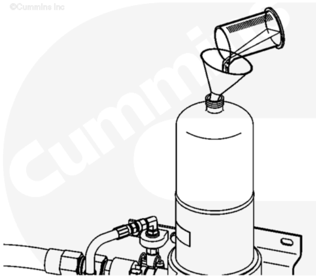
Unscrew the retainer shaft from the threaded boss in the top of the filter can.

Remove the vent cap from the top of the can.




**⚠ CAUTION ⚠**

**Do not spill or drain fuel into the bilge area when disconnecting or removing fuel lines, replacing filters, and priming the fuel system. Do not drop or throw filter elements into the bilge area. The fuel must be disposed of in accordance with local environmental regulations.**



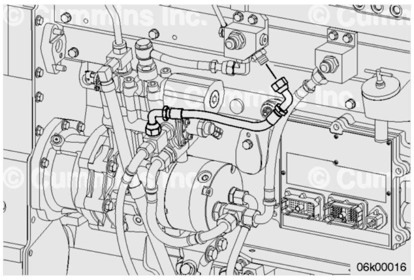

Fill the Stage 1 fuel filter with clean diesel fuel.

Install the vent cap.

Tighten the vent cap by hand **only**.

Open the fuel supply shutoff valve.

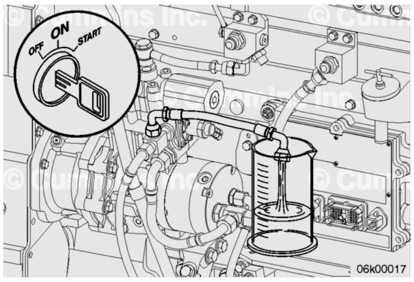
Remove the air bleed line from the drain manifold block and place it into a collection container




Turn the keyswitch ON and observe fuel exiting the air bleed line.

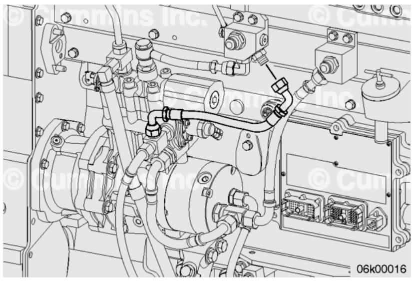
When a steady stream of fuel is observed, turn the keyswitch OFF. If the priming pump stops operating before steady stream of fuel is observed, turn the keyswitch OFF, wait 30 seconds, and turn the keyswitch ON.

Repeat as necessary to observe a steady stream of fuel.



Install an air bleed line to drain the manifold block.

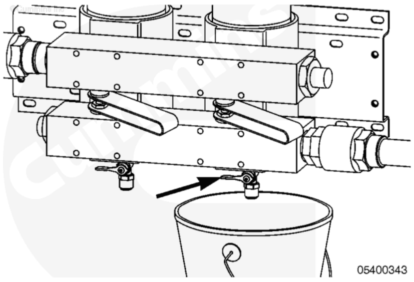
**Torque Value:** 45 n•m [ 33 ft-lb ]



**Note : This procedure explains Classed fuel system filter changes with the engine operating at idle. If the engine Classed, and the intent is to change filters with the engine shut down, follow the Non-Classed procedure.**

## Marine Classed Applications

- Stage 1 and Stage 2 filters **must** be primed prior to starting the engine.
- Procedure for priming Stage 1 and Stage 2 filtration is covered in the same steps, as follows.

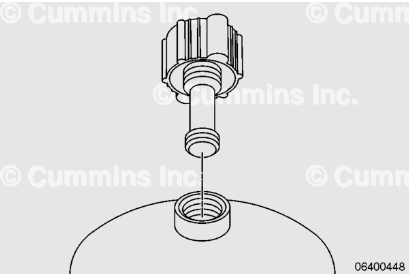


Unscrew the threads of the vent cap.

Lift the vent cap until the retainer shaft catches.

Unscrew the retainer shaft from the threaded boss in the top of the filter can.

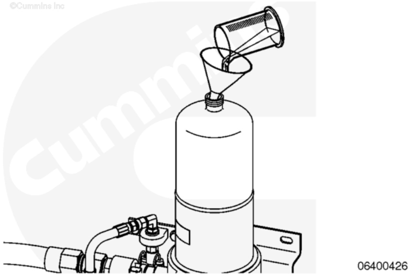
Remove the vent cap from the top of the can.



Fill the filter with clean diesel fuel.

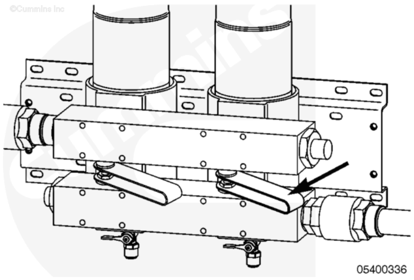
Install the vent cap.

Tighten the vent cap by hand **only**.



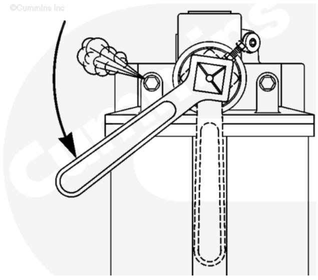
Slowly open the valve handle to the ON position, allowing extra time to purge air from the system.

Repeat the procedure for the remaining filter replacement.



Open the filter vent and slowly move the selector handle toward the center position, allowing fuel to fill the filter that was changed.

Once fuel starts to be discharged from the air vent, close the air vent and move the T-handle to the center position.



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**Torque Value:**
Air Vent 11 n•m [ 97 in-lb ]

Repeat the procedure for the remaining filter, as both filters **must** be used during normal operation.

# Finishing Steps

## with Electronically Actuated Injector

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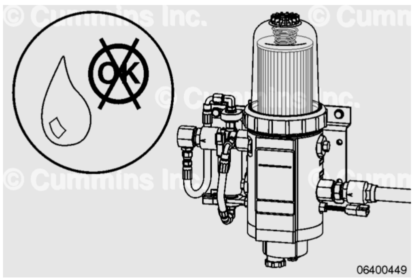


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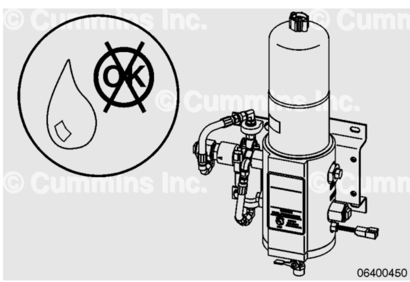
### Industrial and Power Generation Applications

- Industrial Pro™ filter head **must** be checked for leaks.
- Check plumbing to and from the Stage 1 and Stage 2 filter heads.
- Operate the engine for 1 to 2 minutes.
- Check for leaks.

### Marine Non-Classed Applications

- Sea Pro™ filter head **must** be checked for leaks.
- Check plumbing to and from Stage 1 and Stage 2 filter heads.
- Operate the engine for 1 to 2 minutes.
- Check for leaks.

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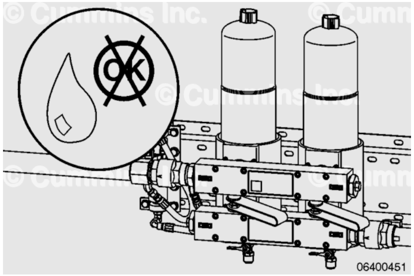
### Marine Classed Applications

- Check the Sea Pro™ filter head for leaks while the engine is operating at idle.
- Check the plumbing to and from the Stage 1 and Stage 2 filter heads.
- Operate the engine for 1 to 2 minutes above idle or at reduced capacity.
- Check for leaks.




Last Modified: 17-Jan-2014